R84 | Four locked models: VAL and TEST separately

Split	Arm	Model Top1 (%)	mean_cost	Actual regret (%)
VAL	A_baseline	71.0852	11.416611477	0.403494
VAL	B_attention_direct	71.0000	11.416697456	0.404999
VAL	C_focal	71.0235	11.417001853	0.408296
VAL	D_attention_focal	70.9543	11.417063866	0.408097
TEST	A_baseline	71.2358	11.373843044	0.394847
TEST	B_attention_direct	71.2235	11.373993105	0.394592
TEST	C_focal	71.2716	11.374011311	0.397181
TEST	D_attention_focal	71.2173	11.373922018	0.395155

ALL = equal weight across18 tasks. Mixed raw mean_cost is descriptive.

Native/FP64 winners. One VAL-selected checkpoint per arm. No TEST-driven rule/checkpoint changes.

Previously published TEST rows have influenced research. These are not fully blind final tests.

Acceptance below uses unrounded VAL values only. The continuing research goal remains unresolved.

R84 | Paired acceptance gates versus A_baseline

Arm	VAL dTop1 (pp)	VAL dCost	VAL dRegret (pp)	VAL accepted	TEST dTop1 (pp)
B_attention_direct	-0.085185	0.000085979	0.001505	False	-0.012346
C_focal	-0.061728	0.000390376	0.004802	False	0.035802
D_attention_focal	-0.130864	0.000452389	0.004603	False	-0.018519

VAL acceptance: Top1 strictly improves, mean_cost does not increase, actual regret does not increase.

All comparisons use the recorded checkpoints; this report does not select a winning deployment.

A smaller number of zero-selected methods is not an acceptance criterion.

R84 | Frozen A-VAL zero / near-zero method targets

Split	Arm	Winners	Recall (%)	Rescued / lost	New false picks	New-FP regret (pp)
VAL	A_baseline	4625	0.24	0 / 0	0	0.000000
VAL	B_attention_direct	4625	0.69	21 / 0	22	0.001277
VAL	C_focal	4625	0.45	10 / 0	17	0.001660
VAL	D_attention_focal	4625	1.21	46 / 1	56	0.004474
TEST	A_baseline	4598	0.28	0 / 0	0	0.000000
TEST	B_attention_direct	4598	0.67	22 / 4	23	0.001178
TEST	C_focal	4598	0.52	13 / 2	15	0.000723
TEST	D_attention_focal	4598	1.13	42 / 3	62	0.003420

Frozen definition: A locked VAL chosen share <=0.1%, with at least one native winner.

This baseline-dependent target list is descriptive/posthoc; it never changes TRAIN sampling.

TEST uses exactly the frozen method/task list. Rescues and false picks are both reported.

R84 | VAL: B_attention_direct versus A_baseline

Task	dTop1 (pp)	Corrected	Harmed	Both wrong: cheaper/dearer	dRegret (pp)
TSP	-0.667	92	122	42 / 53	0.00117
CVRP	-0.133	170	176	151 / 164	0.01921
ATSP	-0.756	99	133	8 / 4	0.00501
OVRP	0.489	271	249	100 / 98	0.00394
VRPB	0.000	23	23	2 / 0	0.02579
VRPL	-1.400	182	245	145 / 132	0.00366
VRPTW	0.067	10	7	0 / 1	-0.00139
OVRPTW	0.022	2	1	0 / 0	-0.00036
OVRPB	0.000	4	4	0 / 0	-0.00106
OVRPL	0.756	236	202	110 / 92	-0.01189
VRPBL	-0.244	18	29	2 / 4	0.01883
VRPBTW	0.089	14	10	0 / 0	-0.00666
VRPLTW	0.067	8	5	0 / 0	-0.00569
OVRPBL	0.022	4	3	1 / 0	-0.00854
OVRPBTW	0.111	9	4	0 / 0	-0.00452
OVRPLTW	0.000	5	5	0 / 0	0.00053
VRPBLTW	0.089	13	9	0 / 0	-0.01104
OVRPBLTW	-0.044	6	8	0 / 0	0.00011

Negative dRegret means lower selected cost. Per-task raw deltas are available in the paired CSV.

Corrected/harmed use native-winner identity, even if exact-cost ties have zero cost change.

R84 | VAL: C_focal versus A_baseline

Task	dTop1 (pp)	Corrected	Harmed	Both wrong: cheaper/dearer	dRegret (pp)
TSP	0.200	78	69	52 / 33	0.00514
CVRP	-0.289	114	127	121 / 142	0.03117
ATSP	-0.244	33	44	1 / 0	0.00708
OVRP	0.356	220	204	82 / 69	-0.00188
VRPB	-0.178	11	19	0 / 2	0.02248
VRPL	-0.556	149	174	107 / 96	0.00751
VRPTW	0.089	6	2	0 / 0	-0.00277
OVRPTW	0.000	1	1	0 / 0	0.00209
OVRPB	-0.044	0	2	0 / 0	0.01358
OVRPL	-0.600	276	303	141 / 130	0.00311
VRPBL	-0.044	9	11	1 / 1	0.00121
VRPBTW	0.133	10	4	0 / 0	-0.00437
VRPLTW	-0.022	3	4	0 / 0	0.00715
OVRPBL	-0.022	2	3	1 / 0	-0.00498
OVRPBTW	0.000	2	2	0 / 0	-0.00018
OVRPLTW	0.022	5	4	0 / 1	0.00450
VRPBLTW	0.067	8	5	0 / 0	-0.00014
OVRPBLTW	0.022	3	2	0 / 0	-0.00427

Negative dRegret means lower selected cost. Per-task raw deltas are available in the paired CSV.

Corrected/harmed use native-winner identity, even if exact-cost ties have zero cost change.

R84 | VAL: D_attention_focal versus A_baseline

Task	dTop1 (pp)	Corrected	Harmed	Both wrong: cheaper/dearer	dRegret (pp)
TSP	0.111	108	103	47 / 48	0.00746
CVRP	-0.156	182	189	168 / 200	0.03579
ATSP	0.244	69	58	4 / 4	-0.00809
OVRP	-0.467	346	367	142 / 124	0.01973
VRPB	-0.089	34	38	3 / 1	0.04536
VRPL	-1.467	237	303	202 / 181	0.00839
VRPTW	0.089	11	7	0 / 1	-0.00203
OVRPTW	0.022	4	3	0 / 0	-0.00264
OVRPB	0.022	7	6	0 / 2	0.00185
OVRPL	-1.022	367	413	180 / 156	-0.00098
VRPBL	-0.200	25	34	3 / 3	0.01094
VRPBTW	0.111	14	9	0 / 0	-0.00261
VRPLTW	0.022	9	8	0 / 0	-0.00342
OVRPBL	-0.044	10	12	2 / 1	0.01241
OVRPBTW	0.133	7	1	0 / 0	-0.02418
OVRPLTW	0.067	9	6	0 / 1	0.00144
VRPBLTW	0.200	13	4	0 / 0	-0.01045
OVRPBLTW	0.067	7	4	0 / 0	-0.00614

Negative dRegret means lower selected cost. Per-task raw deltas are available in the paired CSV.

Corrected/harmed use native-winner identity, even if exact-cost ties have zero cost change.

R84 | TEST: B_attention_direct versus A_baseline

Task	dTop1 (pp)	Corrected	Harmed	Both wrong: cheaper/dearer	dRegret (pp)
TSP	-0.133	98	104	40 / 62	-0.00988
CVRP	0.044	191	189	152 / 162	0.01834
ATSP	-0.111	115	120	11 / 4	-0.00467
OVRP	-0.044	252	254	110 / 95	-0.01757
VRPB	-0.333	19	34	2 / 2	0.03140
VRPL	-0.156	194	201	148 / 128	-0.00959
VRPTW	0.044	12	10	1 / 0	-0.00243
OVRPTW	-0.067	3	6	0 / 0	-0.00014
OVRPB	0.111	7	2	1 / 0	-0.01470
OVRPL	0.600	249	222	117 / 88	-0.01441
VRPBL	-0.089	25	29	1 / 2	-0.00166
VRPBTW	-0.133	10	16	0 / 0	0.01546
VRPLTW	-0.178	2	10	1 / 0	0.00620
OVRPBL	-0.089	0	4	0 / 0	0.00808
OVRPBTW	0.000	2	2	0 / 0	0.00134
OVRPLTW	0.044	5	3	0 / 0	-0.00281
VRPBLTW	0.222	22	12	0 / 0	-0.00982
OVRPBLTW	0.044	5	3	0 / 0	0.00225

Negative dRegret means lower selected cost. Per-task raw deltas are available in the paired CSV.

Corrected/harmed use native-winner identity, even if exact-cost ties have zero cost change.

R84 | TEST: C_focal versus A_baseline

Task	dTop1 (pp)	Corrected	Harmed	Both wrong: cheaper/dearer	dRegret (pp)
TSP	0.289	79	66	54 / 41	0.00232
CVRP	0.800	152	116	95 / 125	0.02932
ATSP	-0.333	30	45	2 / 0	0.01890
OVRP	0.622	219	191	94 / 94	-0.01460
VRPB	0.133	15	9	0 / 1	0.00797
VRPL	0.000	163	163	131 / 85	-0.00830
VRPTW	0.022	5	4	0 / 0	-0.00247
OVRPTW	0.022	2	1	0 / 0	-0.00034
OVRPB	0.022	3	2	1 / 0	-0.00394
OVRPL	-0.644	300	329	149 / 117	-0.00273
VRPBL	0.022	21	20	0 / 2	-0.00723
VRPBTW	-0.178	3	11	0 / 0	0.00442
VRPLTW	-0.111	4	9	1 / 0	0.01540
OVRPBL	-0.044	1	3	0 / 0	0.00247
OVRPBTW	-0.067	1	4	0 / 0	0.00721
OVRPLTW	0.067	4	1	0 / 0	-0.00139
VRPBLTW	0.044	12	10	0 / 0	-0.00764
OVRPBLTW	-0.022	3	4	0 / 0	0.00263

Negative dRegret means lower selected cost. Per-task raw deltas are available in the paired CSV.

Corrected/harmed use native-winner identity, even if exact-cost ties have zero cost change.

R84 | TEST: D_attention_focal versus A_baseline

Task	dTop1 (pp)	Corrected	Harmed	Both wrong: cheaper/dearer	dRegret (pp)
TSP	0.600	107	80	44 / 53	-0.00152
CVRP	0.756	233	199	162 / 161	0.00471
ATSP	0.289	66	53	11 / 2	-0.01416
OVRP	-0.600	348	375	152 / 147	-0.00841
VRPB	-0.422	25	44	1 / 4	0.04513
VRPL	-0.044	268	270	215 / 190	-0.00838
VRPTW	0.000	10	10	0 / 0	-0.00245
OVRPTW	-0.067	5	8	0 / 0	-0.00091
OVRPB	-0.022	10	11	2 / 0	-0.00476
OVRPL	-0.778	396	431	190 / 175	-0.00964
VRPBL	-0.089	35	39	0 / 3	-0.00723
VRPBTW	-0.067	10	13	0 / 0	0.01263
VRPLTW	-0.156	3	10	1 / 1	0.00547
OVRPBL	-0.067	8	11	1 / 1	0.00813
OVRPBTW	0.000	2	2	0 / 0	0.00441
OVRPLTW	0.089	6	2	0 / 0	-0.00147
VRPBLTW	0.222	23	13	0 / 0	-0.01384
OVRPBLTW	0.022	6	5	0 / 0	-0.00215

Negative dRegret means lower selected cost. Per-task raw deltas are available in the paired CSV.

Corrected/harmed use native-winner identity, even if exact-cost ties have zero cost change.

R84 | Locked checkpoint identities

Arm	Selected update	SHA256 prefix
A_baseline	37440	23fb9a5cf621045138fcdcfe
B_attention_direct	37440	309b90d7a1bc5470465140e3
C_focal	46080	a7f7626caa770d0dce67c436
D_attention_focal	43200	addbfaff33b5720dfec357a4

Full hashes and prediction hashes: comparison_artifacts/protocol.json.

Same81000/4500/4500 population; fresh shared initialization; seed2; logical batch1024.

This CPU report reads saved predictions. It starts no inference, training, solver or retry.